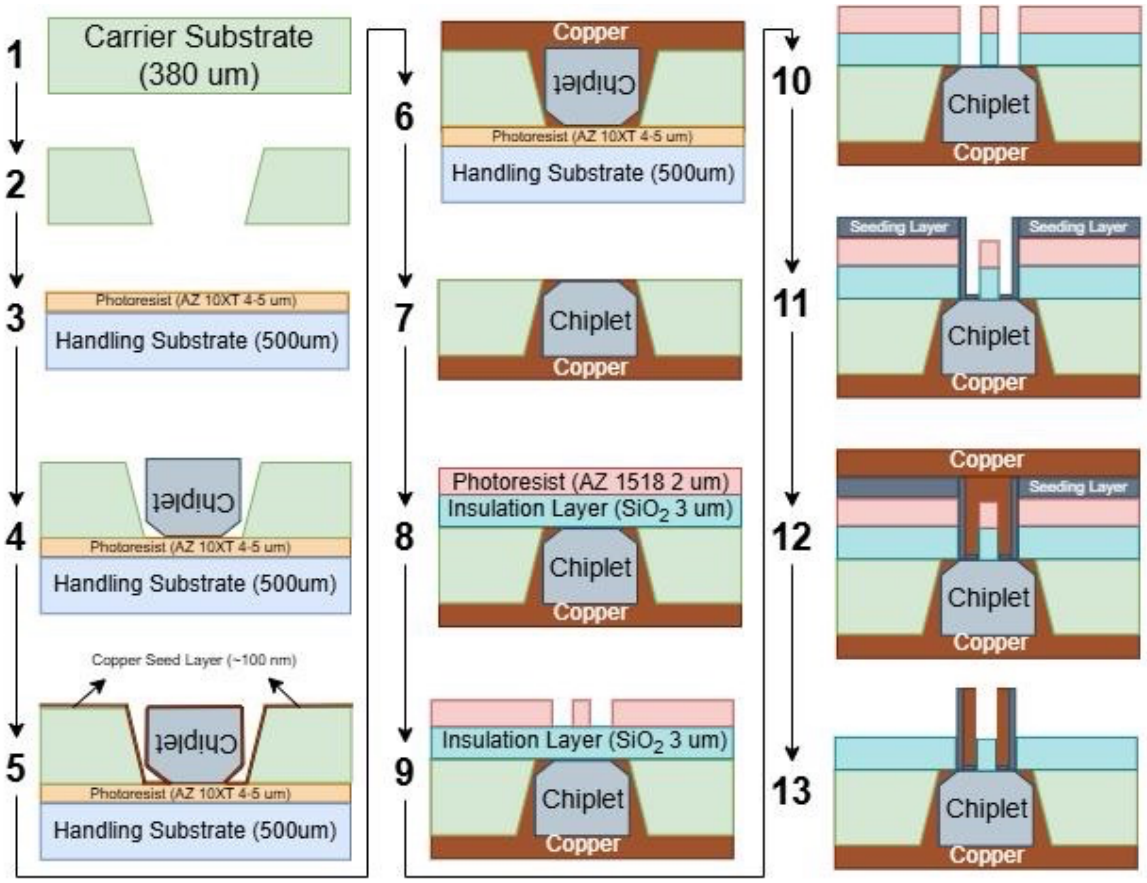

STEPS (Process-flow, In Words):

1. Wafer Cleaning & Preparation (Pre-Coating) Flow Chart Step 1
    a. Initial Rinse
        i. Rinse wafer with DI water
    b. TAI Cleaning
        i. Toluene: 5 min, rinse with DI, dry with nitrogen gun
        ii. Acetone: 5 min, rinse with DI, dry with nitrogen gun
        iii. IPA: 5 min, rinse with DI, dry with nitrogen gun
    c. Dehydration Bake
        i. Pre-heat hot plate to 50˚C, wipe surface with DI water
        ii. Bake wafer at 150˚C for 2 min
        iii. Cool down and blow off dust with nitrogen gun

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2. Photoresist Coating (AZ10XT - 2 Layers) Flow Chart Step 1
    a. First Layer
        i. Mount wafer into spinner
        ii. Dispense AZ10XT at center
        iii. Spin at 1350 rpm for 60s (4s ramp-up)
        iv. Soft bake at 110˚C for 80s
    b. Second Layer

        i. Remount wafer into spinner
        ii. Spin again at 1350 rpm for 60s
        iii. Soft bake at 115˚C for 180s
    c. Wait
        i. Store wafer under yellow light for 35 minutes

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3. Exposure Flow Chart Step 1
    a. Choose appropriate tool:
        i. MJB3: CI2, 405nm, 20mW/cm², 100s
        ii. MJB4: 60mW/cm², 40s
        iii. MA6: CI2, 405nm, 20mW/cm², 100s
            1. Note: Ensure proper mask-wafer contact using the z-axis knob.

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4. Development Flow Chart Step 1
    a. Prepare 1:4 AZ400K:DI water developer solution
    b. Submerge wafer for 12 minutes
    c. Inspect under optical microscope (440nm+ filter) or Alpha-step
    d. Hard bake at 100˚C for 20 minutes

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5. Etching Flow Chart Step 2
    a. Pre-Etch Cleaning
        i. Rinse with DI water for several minutes
        ii. Plasma clean using Tergeo
    b. Wafer Bonding with Crystalbond
        i. Heat 6" wafer to 80˚C
        ii. Apply crystalbond to center (ensure no air bubbles)
        iii. Place 4" wafer on top, heat for 5 min to reflow
    c. Etching Process
        i. Etch for 10 min (10um etch rate)
    d. Post-Etch
        i. Heat to 80˚C to remove 4" wafer
        ii. Strip AZ10XT using acetone, rinse with IPA
        iii. Inspect with Keyence microscope

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6. Preparation of Chiplet and Handling Substrate Flow Chart Step 3
    a. Chiplet Cleaning
        i. Rinse chiplets with IPA only, dry with nitrogen
        ii. Preheat to 50˚C, clean surface
        iii. Bake at 150˚C for 2 min

iv. Cool and dry with nitrogen

b. Handling Substrate Cleaning

i. Prepare Si Wafer 500um

ii. Clean 4" Si wafer with TAI Method

iii. Dry with nitrogen

iv. Preheat to 50°C, clean surface

v. Bake at 150°C for 2 min

vi. Cool and dry with nitrogen

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7. Attaching Chiplet with Carrier Substrate Flow Chart Step 4

a. Adhesive Solution

i. Option A: Photoresist (AZ1518 or AZ10XT)

1. Spin coat at 3500 rpm for 45s (3s ramp)
2. Attach carrier wafer (etched side down) with light pressure
3. Place chiplets (PD and latch) face-down in holes
4. Confirm placement under optical microscope (Keyence 7000)
5. Soft bake at 100°C for 3 min

ii. Option B: Crystalbond 509

1. Heat handling wafer to 125°C
2. Apply crystalbond in center
3. Place carrier wafer + chiplets, heat for 5 min

iii. Option C: Adhesive Tape (Exploratory)

1. Investigate double-sided tape or cleanroom-approved adhesives

b. Chip Alignment with Adhesive-Ready Hole

i. Primitively Poking the Chiplets into the Etched Holes of the Carrier Substrate

1. Use Keyence 7000 to examine alignment
2. Ensure a gap is left between the Carrier Substrate and around the Chiplet.

ii. Explore Less Pathetic Alternative Methods in Chip Alignment

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8. Gap Preparation and Backfilling Gaps for Structural Integrity Flow Chart Steps 5, 6

a. Adhesive Removal in Gaps

i. Photoresist: Mass UV exposure + developer droplets + nitrogen blow dry

ii. Crystalbond: Droplet of 80°C water + dry + repeat

b. Backfilling Gap with PVD (Metallic) Deposition of a Cu Seeding Layer

i. Deposit a ~100nm Copper Seeding Layer

c. Growing the Cu Seeding Layer with Electroplating

i. 80mA Forward Current - 10s

ii. 160mA Reverse Current - 1s (for flat coating)
iii. Examine the Gap Height with P7-Profilometer and/or Keyence 7000 to ma ke sure copper covers Carrier Substrate

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9. Handling Wafer Separation Flow Chart Step 7
    a. Soak in Acetone or 80°C water bath for 30+ min (based on adhesive)
    b. Gently separate and clean residue.
    c. Examine peeled side with P7-Profilometer and/or Keyence 7000 to see if structural integrity is affected.

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    d. Pre-Metallization Preparation Flow Chart Steps 8, 9, 10, 11
        i. Dehydration Bake at 85°C for 1 min
        ii. Make sure the copper side is on the bottom, and the chiplet interconnect si de is on the top.
        iii. Deposit 3um SiO2 insulation with HPCVD (Apex SLR) or PVD.
        iv. Deposit 2um AZ10XT.
            1. Dispense PR (20 pipettes)
            2. Spin at 3000 rpm for 60s
            3. Soft bake at 110°C for 120s
            4. Expose using metallization mask
            5. Wait 30 – 45 min (post-exposure bake delay)
            6. Develop Photoresist with a development solution at 1:4 ratio of AZ 400K to DI water for 12mins.
            7. Hard bake at 95°C for 20 min
        v. Conduct exposure and etching on the AZ10XT surface to get the desired p attern.
            1. Examine peeled side with P7-Profilometer and/or Keyence 7000.
        vi. Conduct exposure and etching on the SiO2 surface with Apex SLR to get t he desired pattern.
            1. Examine peeled side with P7-Profilometer and/or Keyence 7000.

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10. Metallization of Interconnects Flow Chart Step 12
    a. Deposit ~100nm of Cu with PVD (Metallic) as Seeding Layer
    b. Growing the Cu Seeding Layer with Electroplating
        i. 80mA Forward Current - 10s
        ii. 160mA Reverse Current - 1s (for flat coating)
    c. Examine the Gap Height with P7-Profilometer and/or Keyence 7000 to make sure copper covers Carrier Substrate

11. Lift-Off of Photoresist and the Layers Above It Flow Chart Step 13
    a. Soak the entire thing under Acetone-DI water solution of 1:4 for 12 mins (subject to change)
    b. The AZ10XT layer would peel off and the layers above it will also peel off
    c. Blow dry with Nitrogen gun
    d. Dehydration Bake 95˚C for 60s (subject to change)
    e. Examine structure under P7-Profilometer and/or Keyence 7000